🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.NOS.9

Determine Distance on the Coordinate Plane

Lesson 7-6 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes (Level 2 Standard)

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the distance between two points on the coordinate plane using absolute value.
Language Objective: I can explain my work using the words distance, absolute value, horizontal distance, and vertical distance.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Distance on the Coordinate Plane | Distance | Absolute value | Horizontal distance | Vertical distance | Coordinate plane | Integer

👆 *Tap any word to see what it means and a picture.*

1. When two points share a coordinate, the distance is the ______ ______ of the difference.

2. How far apart two points are on the coordinate plane is the ______.

3. The distance of a number from zero, used to find distance between points, is its ______.

4. The distance between two points along the x-direction is the ______.

5. The distance between two points along the y-direction is the ________.

6. The grid made by the x-axis and y-axis is the ________.

7. A whole number or its opposite is an ________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How does the drone use coordinate distance to calculate the total flight path?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Distance on the Coordinate Plane** — The positive space between points, found from the absolute difference of coordinates on a horizontal or vertical line.
  Distancia en el plano de coordenadas — El espacio positivo entre puntos, que se halla con la diferencia absoluta de las coordenadas en una línea horizontal o vertical.
- [ ] **Distance** — How far apart two points are. It is never negative.
  Distancia — Qué tan separados están dos puntos. Nunca es negativo.
- [ ] **Absolute value** — How far a number is from zero. It is never negative.
  Valor absoluto — Qué tan lejos está un número de cero. Nunca es negativo.
- [ ] **Horizontal distance** — How far apart two points are going left or right.
  Distancia horizontal — Qué tan separados están dos puntos yendo a la izquierda o a la derecha.
- [ ] **Vertical distance** — How far apart two points are going up or down.
  Distancia vertical — Qué tan separados están dos puntos yendo hacia arriba o hacia abajo.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The horizontal distance from (-4, 0) to (3, 0) is ____ units because ____. The vertical distance from (3, 0) to (3, -5) is ____ units because ____. The total flight path is ____ units.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: I can find the distance by looking at the coordinates that ____.

Evidence: Student finds the distance by subtracting the x-coordinates (7 - 2 = 5) since the points share a horizontal line.

Reasoning: This evidence connects the definition of distance on the coordinate plane to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The horizontal distance from (-4, 0) to (3, 0) is ____ units because ____. The vertical distance from (3, 0) to (3, -5) is ____ units because ____. The total flight path is ____ units.**
- **My answer is ____.**

  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

  Mi afirmación es ____.
- **I know because ____.**

  Lo sé porque ____.
- **So ____.**

  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Distance on the Coordinate Plane
2. **Notes 2:** Distance
3. **Notes 3:** Absolute value
4. **Notes 4:** Horizontal distance
5. **Notes 5:** Vertical distance
6. **Notes 6:** Coordinate plane
7. **Notes 7:** Integer
8. **Watch for this mistake:** *A common mistake in Distance on the Coordinate Plane is subtracting the coordinates as if both points were on the same side of zero, instead of adding their absolute values when the points are on opposite sides. For example, to find the distance between (-6, 3) and (2, 3), a student might subtract 6 − 2 = 4, but since -6 and 2 are on opposite sides of zero, the absolute values must be added instead: |-6| + |2| = 6 + 2 = 8. The correct distance is 8 units, not 4. Before you submit, ask: 'Are these two points on opposite sides of zero — and if so, did I add the absolute values instead of subtracting?'*
9. **Try It 1:** B. 7 blocks — *Same y = 5, so the segment is horizontal. The points are on opposite sides of zero, so add the absolute values: |-4| + |3| = 4 + 3 = 7 blocks.*
10. **Try It 2:** B. 5 blocks — *Same x = -2, so the segment is vertical. The points are on opposite sides of zero: |1| + |-4| = 1 + 4 = 5 blocks.*